

Notes YEAST_COB_Batch constraint-based approach

```
In [1]: # Import the application model and context variables
        from BPL_YEAST_COB_Batch_fmipy import*
```

Windows - run FMU pre-compiled JModelica 2.14

```
In [2]: # Import the general FMU_explore functions and adapt them to the application
        from fmu_explore_fmipy import AdaptWith

        Adapted = AdaptWith(parValue, parLocation, parCheck, keyVariables, \
                               model_description, fmu_model, fmu_process_diagram, \
                               MSL_usage, MSL_version, BPL_version, \
                               opts_std, simulationTime, timeDiscreteStates, \
                               diagrams, ax, lines)

        par = Adapted.par
        init = Adapted.init
        disp = Adapted.disp
        simu = Adapted.simu
        show = Adapted.show

        resetPen = Adapted.resetPen
        process_diagram = Adapted.process_diagram
        describe_general = Adapted.describe_general
        describe_parts = Adapted.describe_parts
        describe_MSL = Adapted.describe_MSL

        FMU_explore_info = Adapted.FMU_explore_info
        system_info = Adapted.system_info
        SDG = Adapted.SDG
```

```
In [3]: run -i BPL_YEAST_COB_Batch_fmipy_explore.py
```

Model for the process has been setup. Key commands:

- par() - change of parameters and initial values
- init() - change initial values only
- simu() - simulate and plot
- newplot() - make a new plot
- show() - show plot from previous simulation
- disp() - display parameters and initial values from the last simulation
- describe() - describe culture, broth, parameters, variables with values/units

Note that both disp() and describe() takes values from the last simulation and the command process_diagram() brings up the main configuration

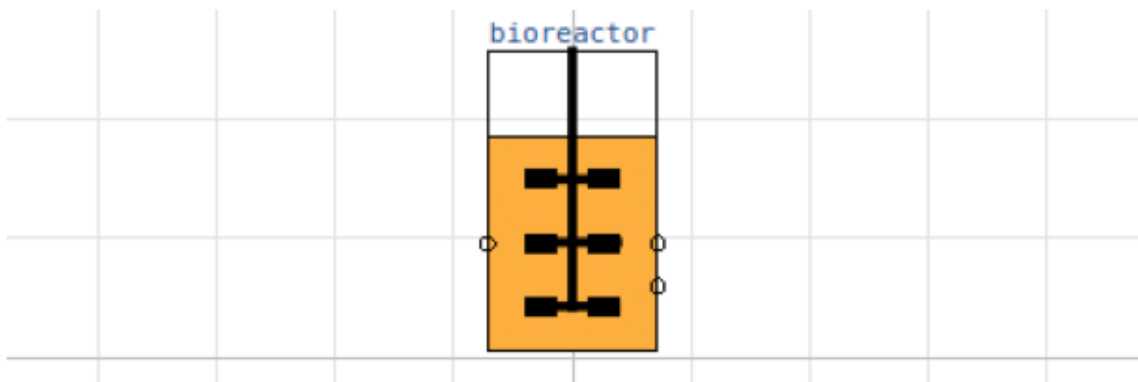
Brief information about a command by help(), eg help(simu)

Key system information is listed with the command system_info()

```
In [4]: plt.rcParams['figure.figsize'] = [20/2.54, 16/2.54]
```

```
In [5]: process_diagram()
```

No processDiagram.png file in the FMU, but try the file on disk.



Try using LP in each step

```
In [6]: from optlang import Model, Variable, Constraint, Objective
```

```
In [7]: # Define culture constraint-based model
def culture(G, E):

    # LP calculation of the optimal qGr, qEr based on G and E values

    # - parameters
    qO2max = 6.9e-3; kog = 2.3; koe = 1.6; YGr = 3.5; YEr = 1.32;
    alpha = 0.01; beta = 1.0

    # - transfer data from dynamic reactor model to static LP model
    qGr_opt = Variable('qGr_opt', lb=0)
    qEr_opt = Variable('qEr_opt', lb=0)

    # - LP model constraint and objective
    mu_max = Objective(YGr*qGr_opt + YEr*qEr_opt, direction='max')
    qO2lim = Constraint(kog*qGr_opt + koe*qEr_opt, ub=qO2max)
    qGlim = Constraint(qGr_opt, ub=alpha*max(0,G))
    qElim = Constraint(qEr_opt, ub=beta*max(0,E))

    # - put together the LP model
    yeast_model = Model(name='Yeast bottleneck model')
    yeast_model.objective = mu_max
    yeast_model.add(qO2lim)
    yeast_model.add(qGlim)
    yeast_model.add(qElim)

    # - do LP optimization
    yeast_model.optimize()

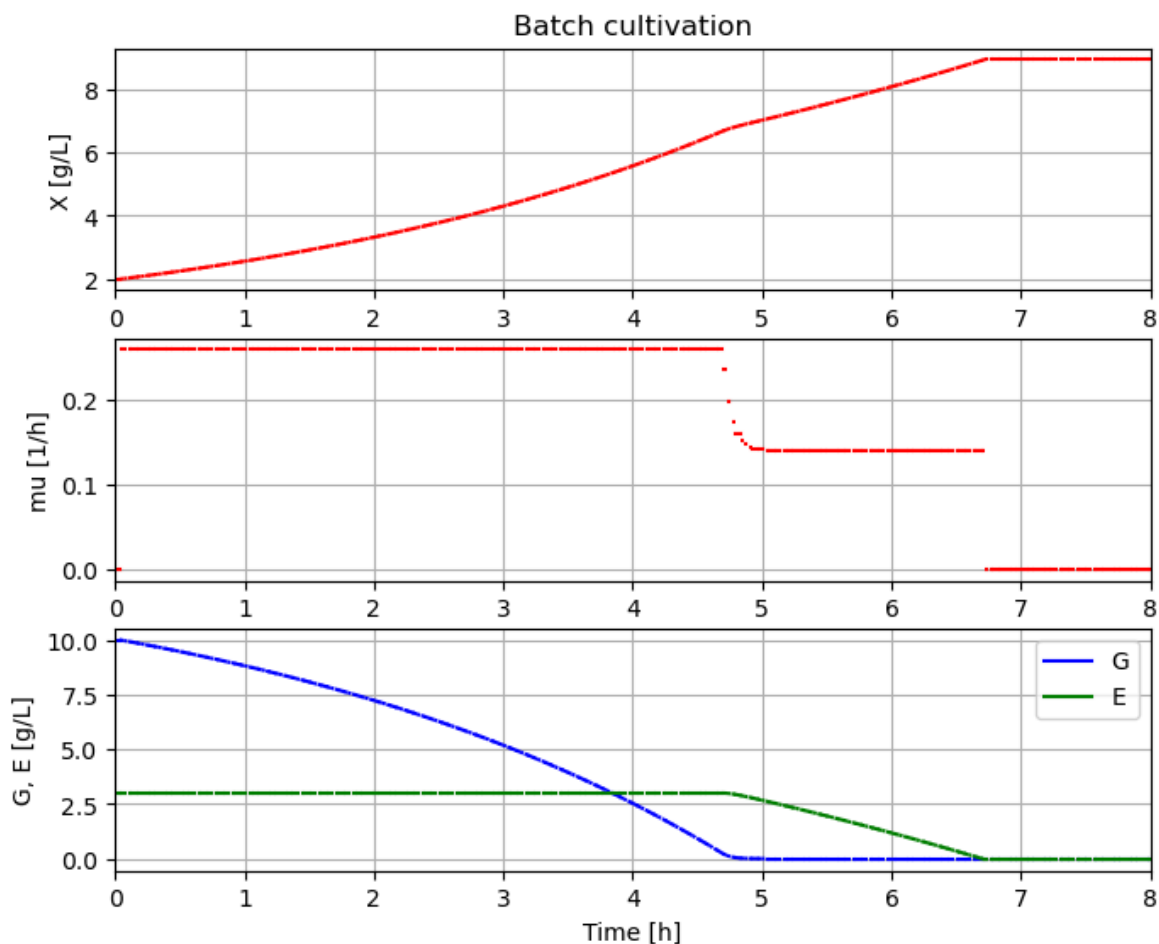
    return (yeast_model.objective.value, yeast_model.variables.qGr_opt.primal, y
```

```
In [8]: # Initialization
V_start=1.0
init(V_start=V_start, VX_start=V_start*2.0, VG_start=V_start*10, VE_start=V_start
```

```
In [9]: # Loop of simulations
t_final = 8.0
t_samp = 0.0333
n_samp = t_final/t_samp + 1
```

```
In [10]: # Simulate n sample steps
newplot(title='Batch cultivation', plotType='TimeSeries2')
ax[0].set_xlim([0, t_final]); ax[1].set_xlim([0, t_final]); ax[2].set_xlim([0, t

simu(t_samp, options=opts_fast)
for i in range(int(n_samp)):
    sim_res = Adapted.sim_res
    (mum_opt, qGr_opt, qEr_opt, qO2_opt) = culture(sim_res['bioreactor.c[2]'][-1
    par(mum=mum_opt, qGr=qGr_opt, qEr=qEr_opt, qO2=qO2_opt)
    simu(t_samp, 'cont', options=opts_fast)
```



```
In [11]: system_info()
```

System information

```
-OS: Windows
-Python: 3.12.11
-Scipy: 1.16.0
-FMPy: 0.3.27
-FMU by: JModelica.org
-FMI: 2.0
-Type: CS
-Name: BPL_YEAST_COB.Batch
-Generated: 2025-07-22T17:39:01
-MSL: 3.2.2 build 3
-Description: Bioprocess Library version 2.3.1
-Interaction: FMU_explore version 1.2.4
```

```
In [12]: !conda list optlang
```

```
# packages in environment at C:\Users\janpa\miniconda3\envs\fmpy0325:
#
# Name                      Version                Build      Channel
optlang                     1.8.3                  pyhd8ed1ab_0  conda-forge

!lsb_release -d
```